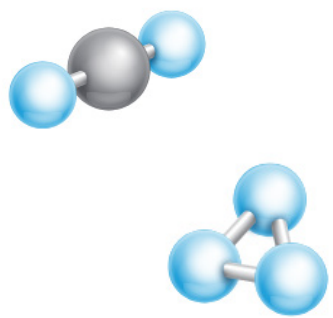
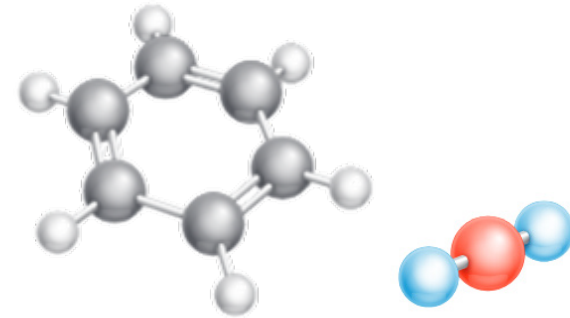
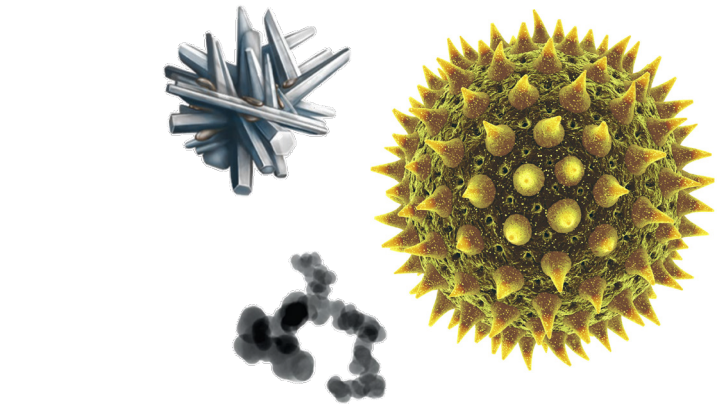


Welcome to AQIQ, Lesson 1!

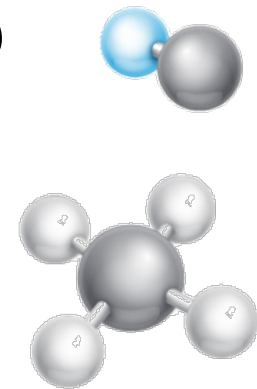
Central AQ Slide Deck

Last Updated August 31, 2026 by Percy Smith

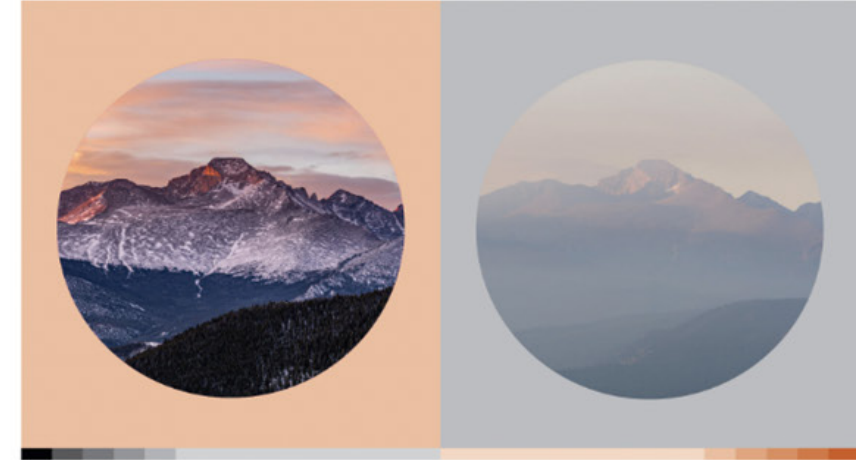




What is Air Quality?

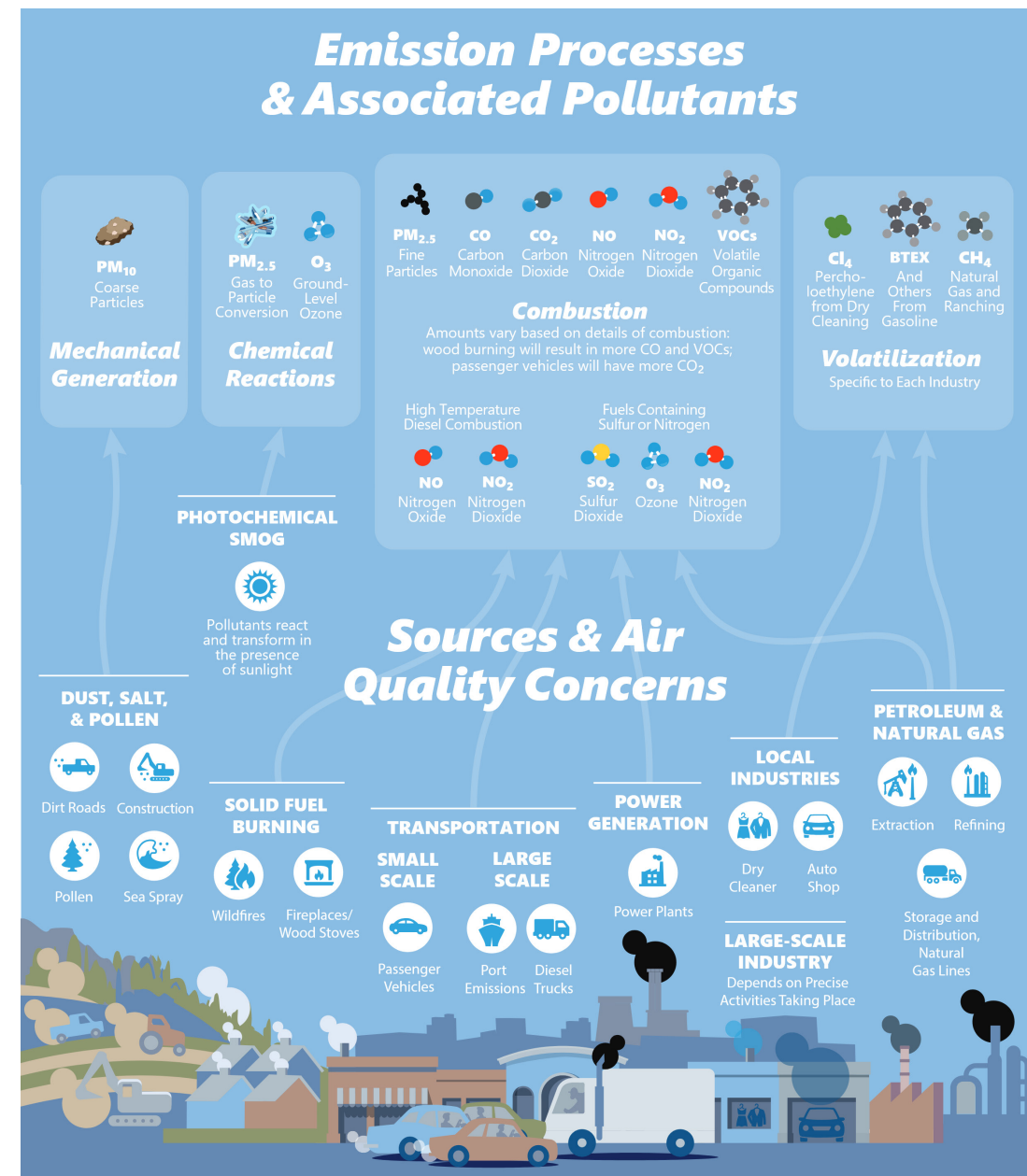


How has the air impacted you?



“Air pollution” is not one thing!

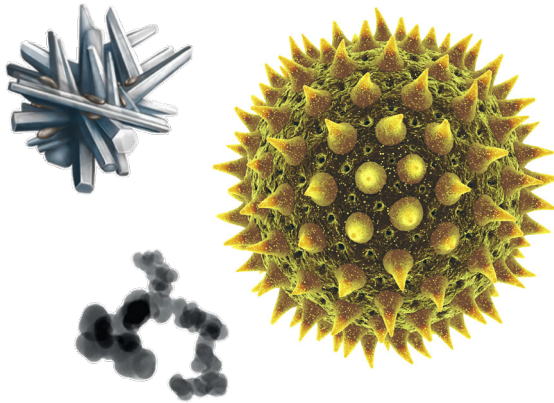
There are many pollutants in the air that impact our health and environment



Types of Air Pollution

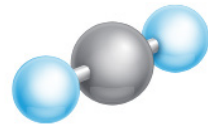
Solids & Liquids

PM

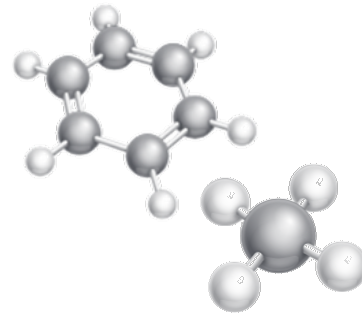


Gases

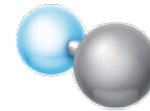
CO₂



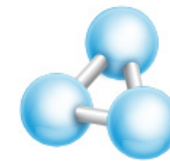
VOC's



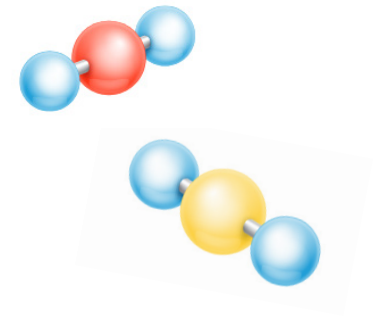
CO



O₃

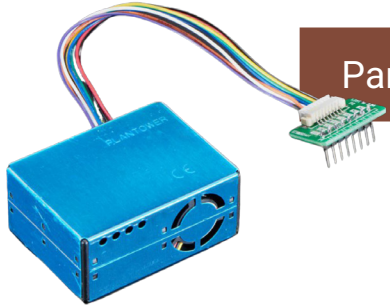


Many more!

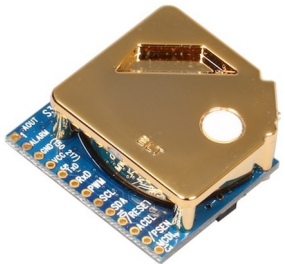




Carbon Monoxide (CO)



Particulate Matter (PM 2.5)



Carbon Dioxide (CO₂)



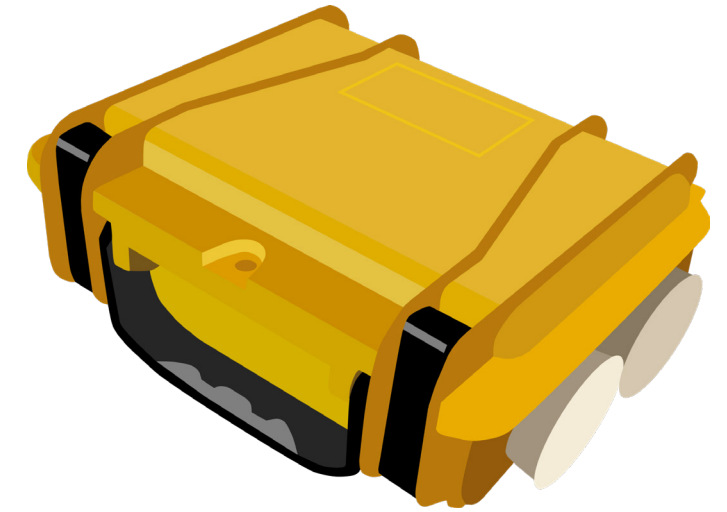
Time recorded in MDT

AQ-IQ Y-Pod Checklist

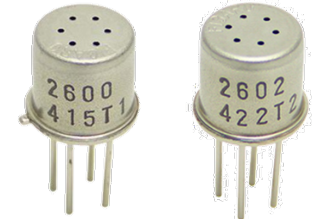
Before you begin collecting data...

- ☐ Is your SD Card inserted?
- ☐ Is your pod plugged into a power source (wall or battery)?
- ☐ Is the left power switch flipped "on"?
- ☐ Are the fans on and spinning?
- ☐ Are the sensors plugged in for the pollutants that you are measuring?
- ☐ Have you marked the time and location when you start recording data?

ALLOW FOR 30 MINS OF WARM UP BEFORE RECORDING DATA



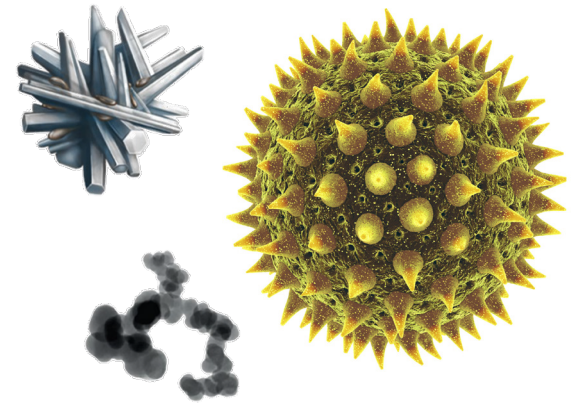
Volatile Organic Compounds (VOC's)



The YPOD

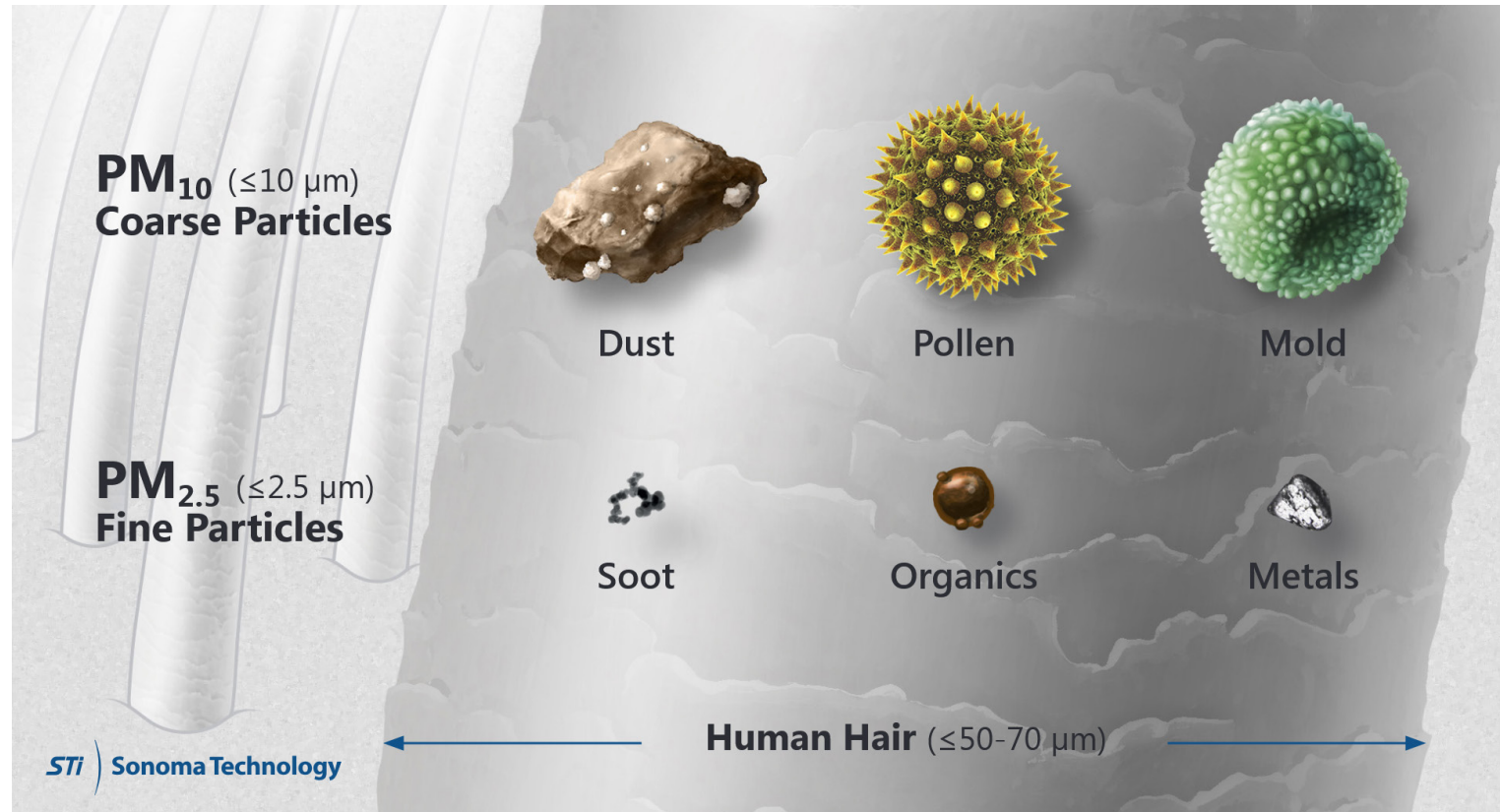


Particulate Matter (PM)

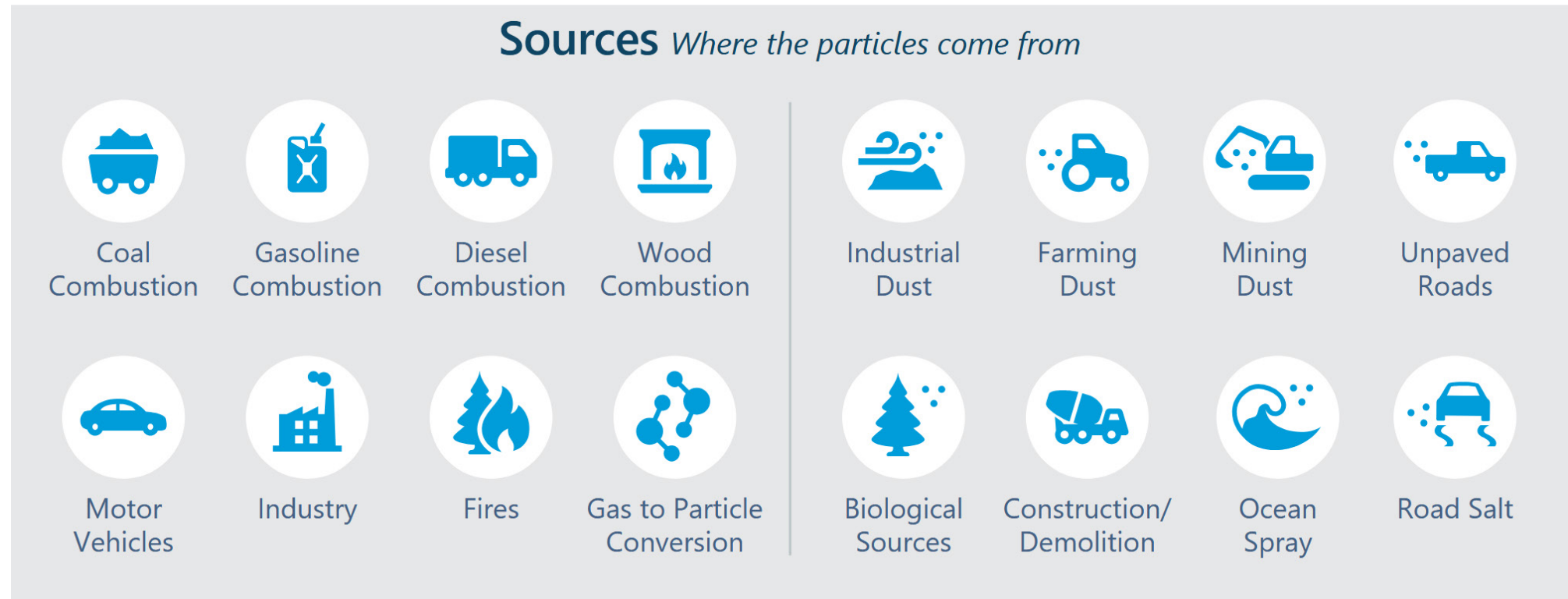


PM: Particulate Matter

- A mixture of solid particles & liquid droplets in the air



Where does PM come from?



Why do we care about PM?



PM₁₀

Coarse particles can irritate the upper respiratory tract.

PM_{2.5}

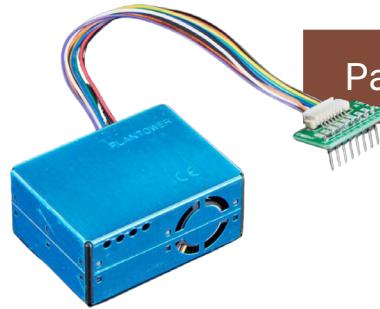
Fine particles can penetrate deep into the lower respiratory tract.

Figure 2-2. Areas of the respiratory tract that particles can reach.



Challenge 1: What will create the largest spike in PM2.5?

- Lighting a match
- Lighting a candle
- Painting nails
- Using nail polish remover
- Drawing with sharpies
- Spraying the mister

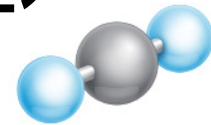


Particulate Matter (PM 2.5)

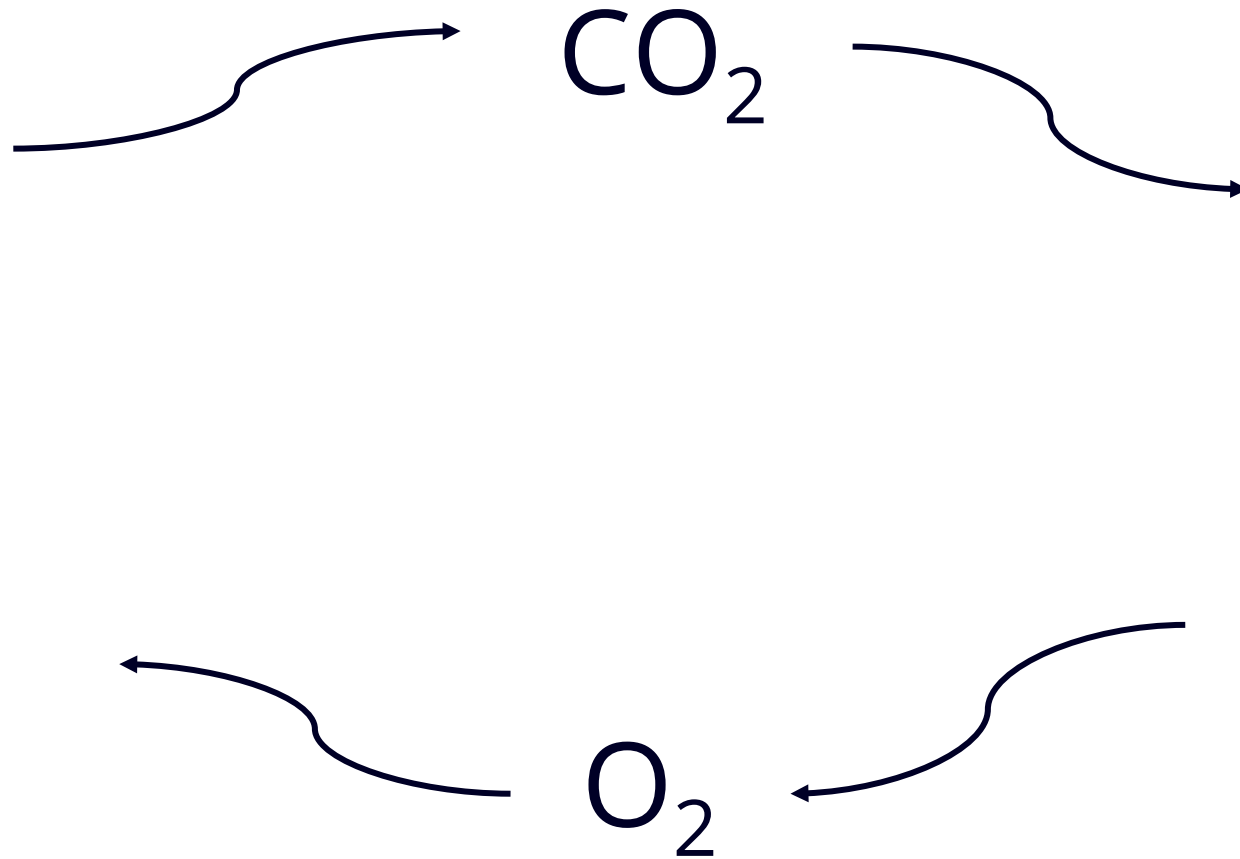




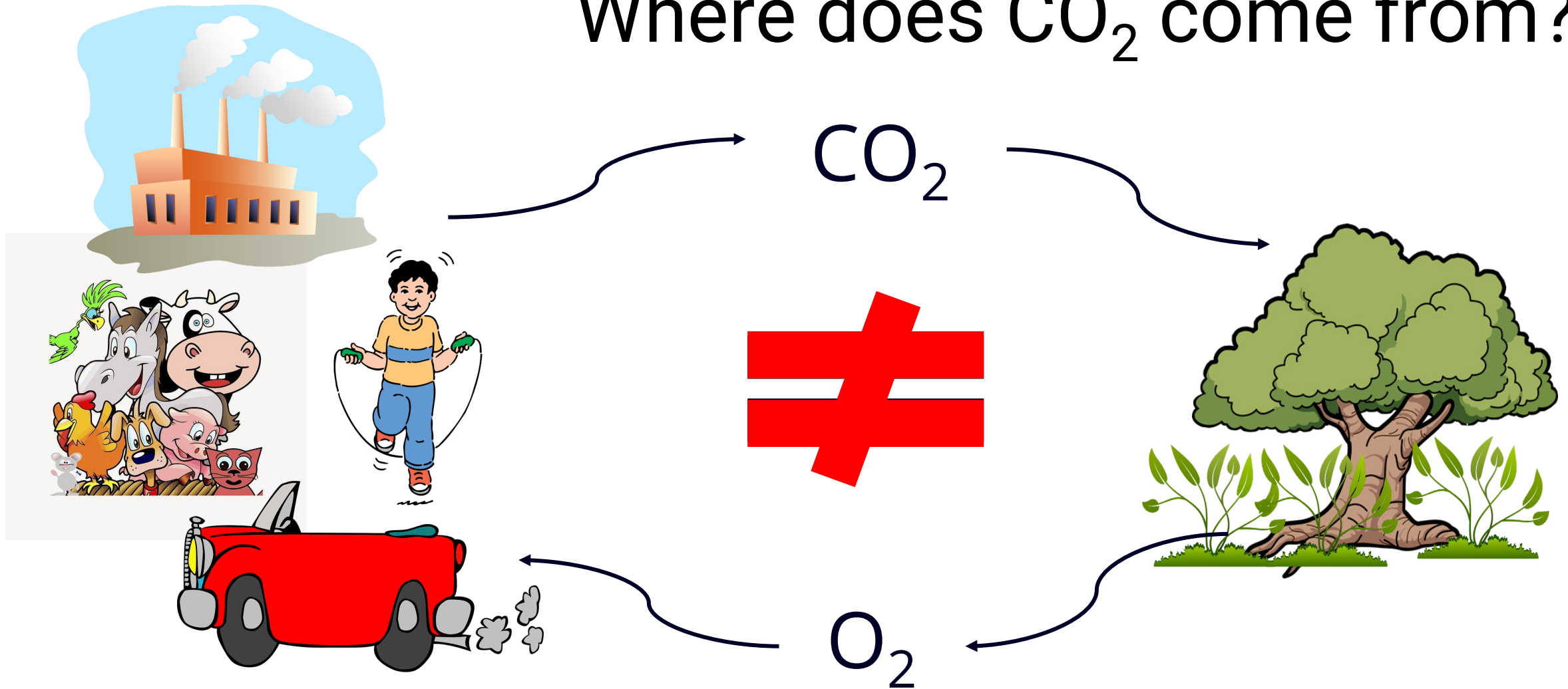
Carbon Dioxide (CO₂)



Where does CO₂ come from?

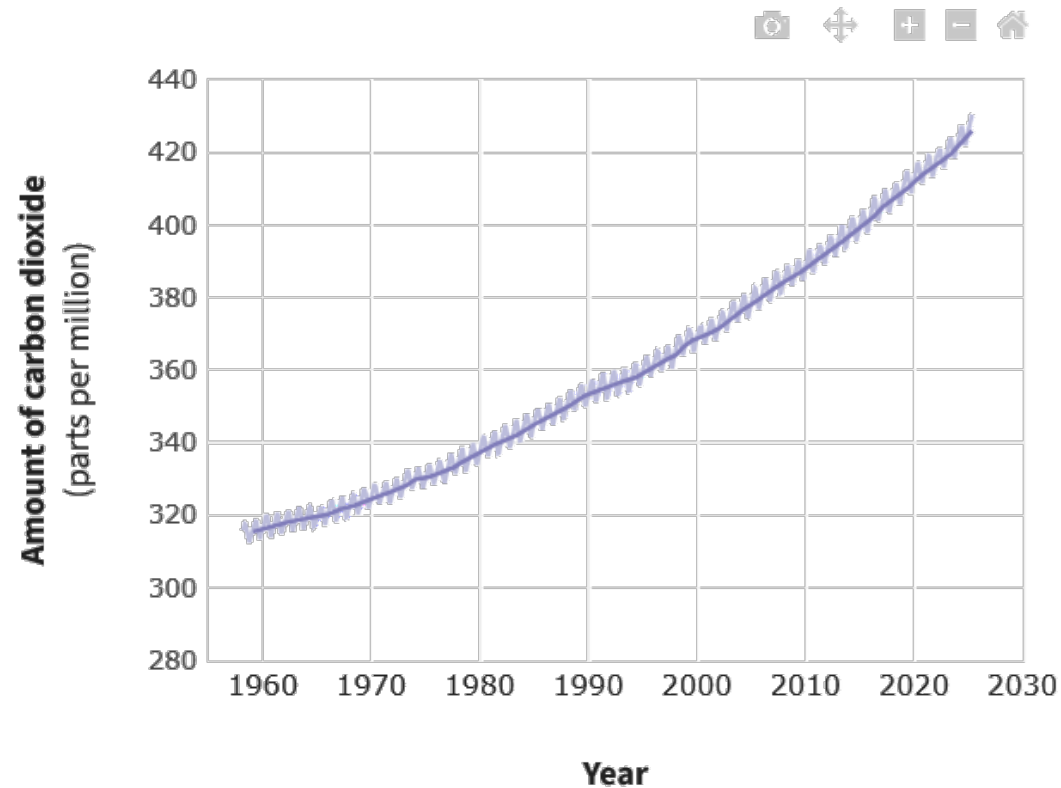


Where does CO₂ come from?

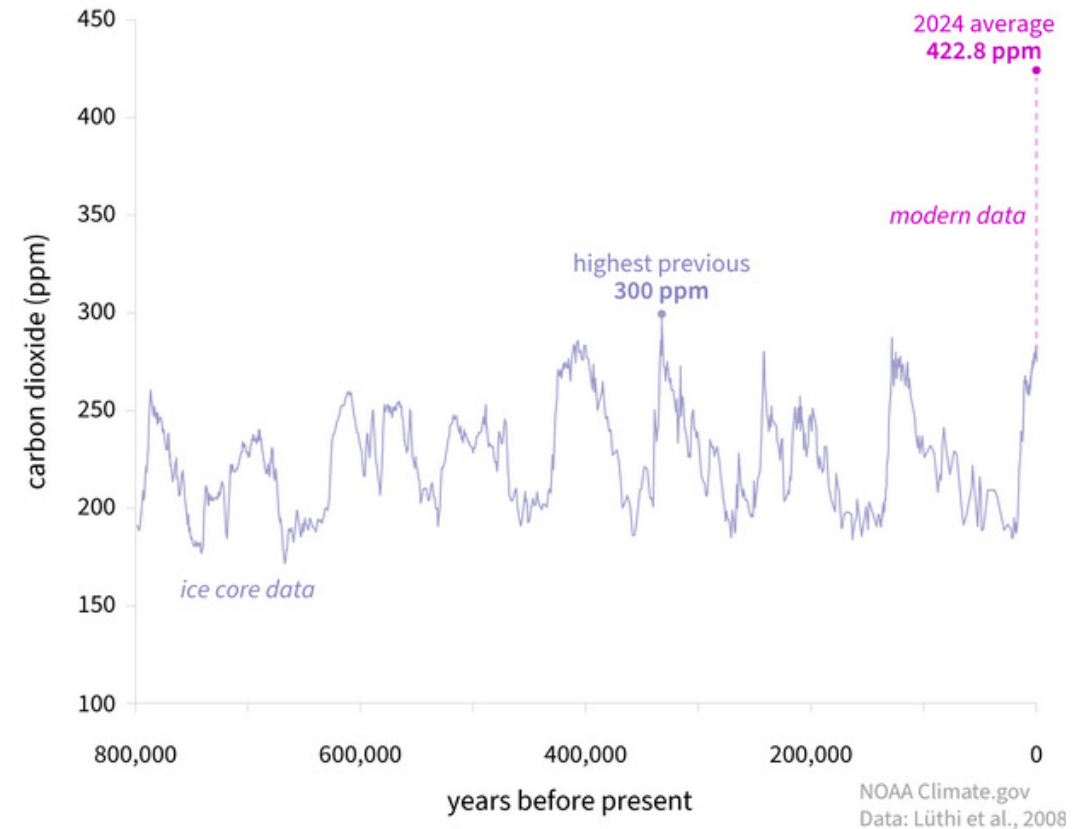


How has CO₂ been changing?

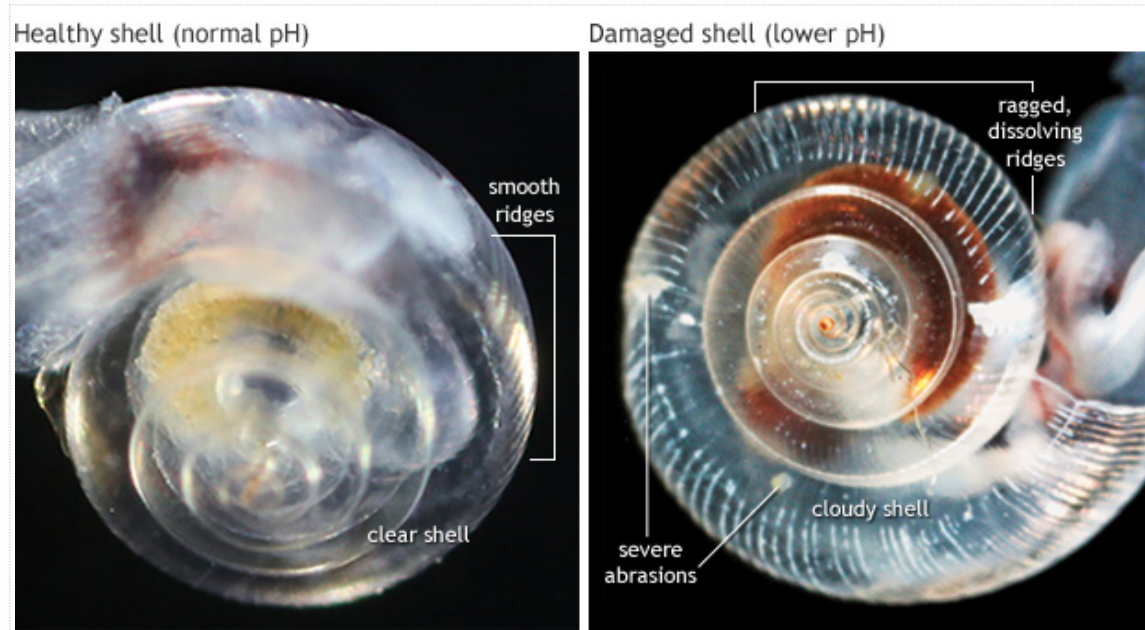
ATMOSPHERIC CARBON DIOXIDE



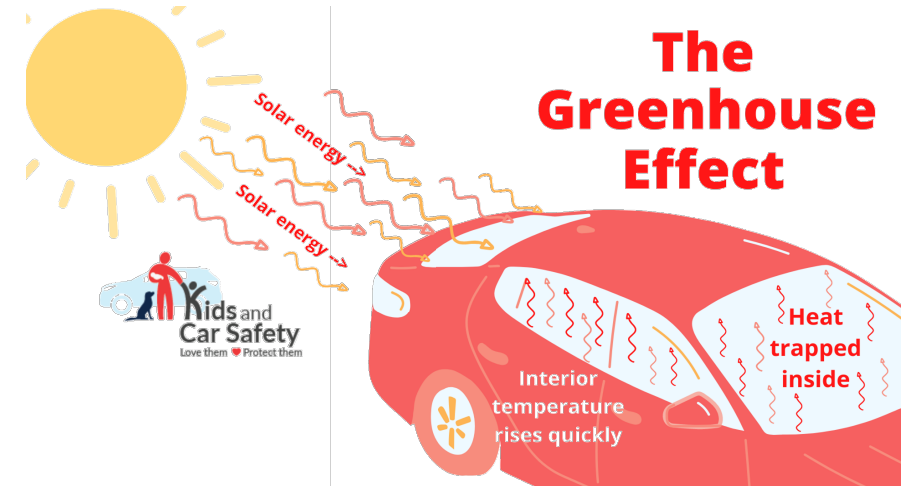
CARBON DIOXIDE OVER 800,000 YEARS



Why does CO₂ matter outside?

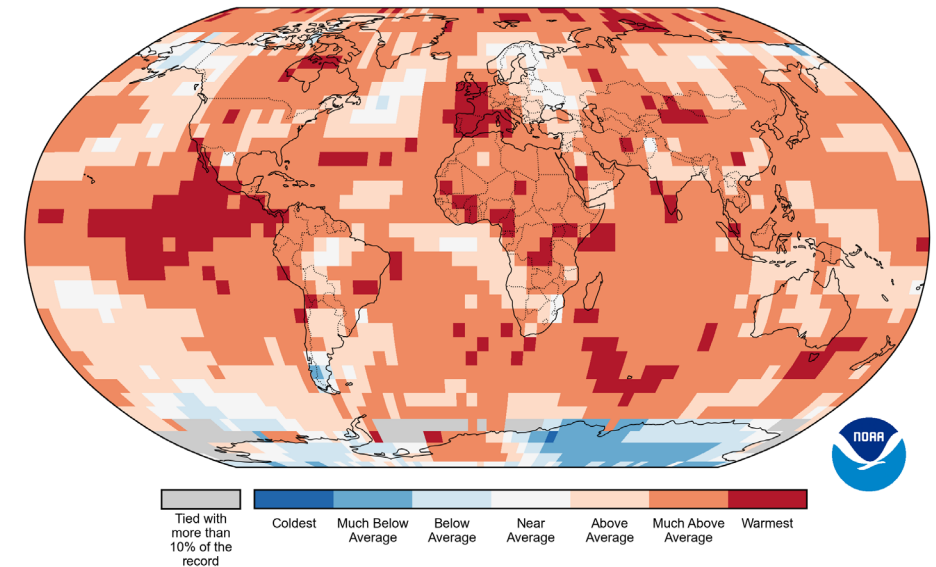


(left) A healthy ocean snail has a transparent shell with smoothly contoured ridges. (right) A shell exposed to more acidic, corrosive waters is cloudy, ragged, and pockmarked with 'kinks' and weak spots. [Photos](#) courtesy Nina Bednarsek, NOAA PMEL.



Jul 2026 Global Temperature Percentiles

NOAA's National Centers for Environmental Information



Source: NOAA GlobalTemp Dataset Version 6.1.0-20260807



Engineering & Applied Science

UNIVERSITY OF COLORADO BOULDER

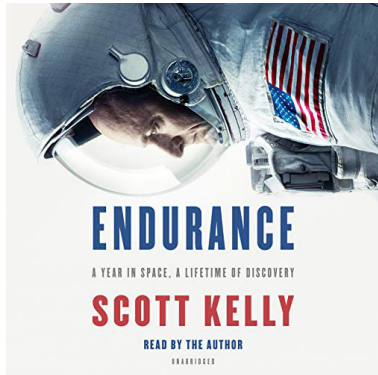
Hot Cars: Charts, Data, and Visuals. (n.d.). Kids and Car Safety. Retrieved August 31, 2026, from <https://www.kidsandcars.org/hot-cars/charts-data-and-visuals>

Lindsey, R. (2025, May 21). Climate change: Atmospheric carbon dioxide. *Understanding Climate* <https://prod-01-asg-www-climate.woc.noaa.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide>

Assessing the Global Temperature and Precipitation in July 2026. (2026). [Summary Report]. National Centers for Environmental Information. <https://www.ncei.noaa.gov/news/global-climate-202607>



Why does CO₂ matter inside?



“The carbon dioxide levels are high today, nearly **four millimeters of mercury**. [...] I can sense the levels with a high degree of accuracy based only on the symptoms I've come to know so well: **headaches, congestion, burning eyes, irritability.**”

(Kelly, 2017, p. 86)

CO₂ concentrations aboard the ISS regularly reach **above 4000 ppm** as of 2024. Research has begun to indicate this is **seriously impacting astronauts' health**.

Review > [Aerosp Med Hum Perform. 2025 Nov;96\(11\):1024-1026. doi: 10.3357/AMHP.6716.2025.](#)

Carbon Dioxide as a Multisystem Threat in Long Duration Spaceflight

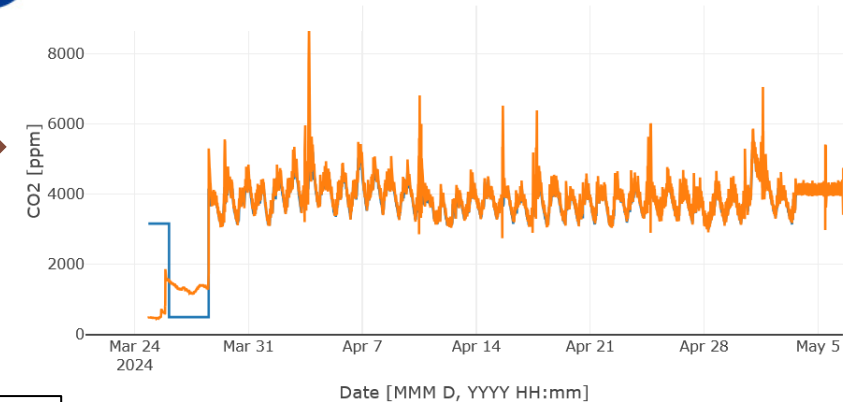
Lorna A Evans

MARCH 1, 2025 | MEDICAL | AEROSPACE | ELECTRONICS & COMPUTERS

Elevated CO₂ Levels on the ISS: Unmasking a Hidden Risk of DNA Damage for Astronauts



Open Science for Life in Space



Engineering & Applied Science

UNIVERSITY OF COLORADO BOULDER

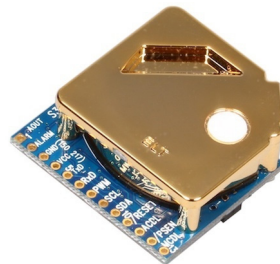
Environmental Data App (RR-28). (2024). [Dataset]. Environmental Data App <https://visualization.osdr.nasa.gov/eda/>
Kelly, S., & Dean, M. L. (2017). *Endurance: A year in space, a lifetime of discovery* (First edition). Alfred A. Knopf.

Stolc, V., Karhanek, M., Griko, Y., & Loftus, D. (2025, March 1). *Elevated CO₂ Levels on the ISS: Unmasking a Hidden Risk of DNA Damage for Astronauts*. 15(3). <https://www.medicaldesignbriefs.com/component/content/article/52628-elevated-co2-levels-on-the-iss-unmasking-a-hidden-risk-of-dna-damage-for-astronauts>

Evans, L. A. (2025). Carbon Dioxide as a Multisystem Threat in Long Duration Spaceflight. *Aerospace Medicine and Human Performance*, 96(11), 1024–1026. <https://doi.org/10.3357/AMHP.6716.2025>

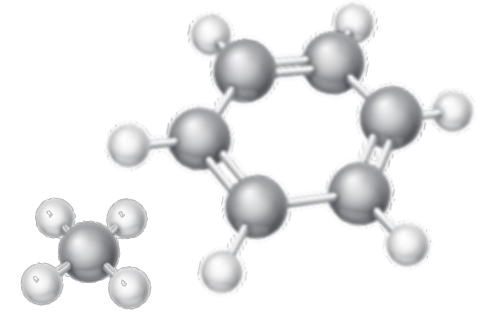


Challenge 2: Whose breath can create the largest spike in CO₂?



Carbon Dioxide (CO₂)





Volatile Organic Compounds (VOCs)



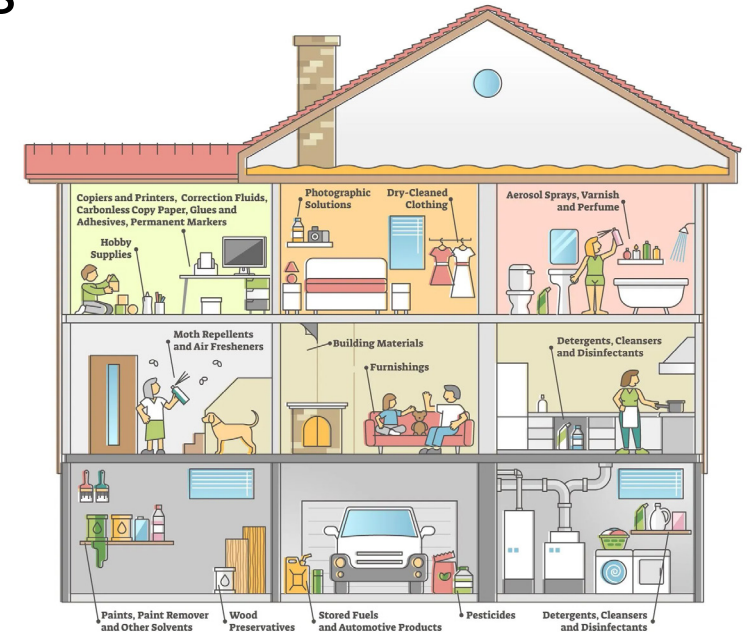
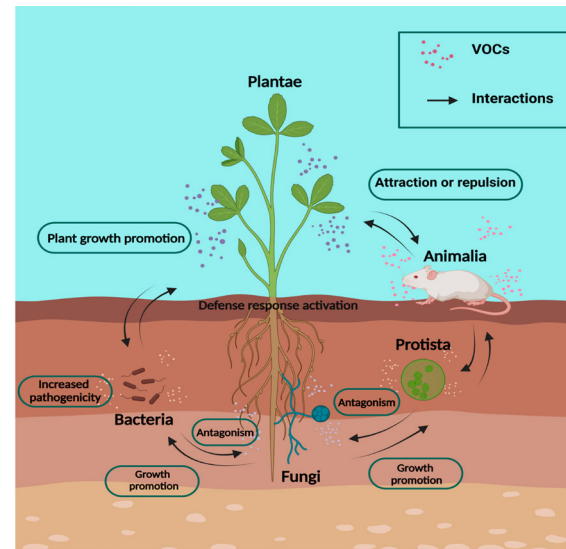
VOC: Volatile Organic Compounds

- Think of it as anything “smelly”
 - Cooking, burning
 - Paints, cleaning products, perfumes, gasoline
 - Smell of pine trees and freshly cut grass



Where do VOC's Come From?

- Indoor & Outdoor Anthropogenic Sources
 - Dry cleaning, industry processes, painting
 - Cleaning products, adhesives, fuels, air fresheners
- Natural Sources
 - Recent research indicates that some plants deploy VOCs for defense



Challenge 3: What will create the largest spike in VOC's?

- Lighting an unscented candle
- Painting nails
- Using nail polish remover
- Drawing with sharpies
- Wiping the table with a wipe



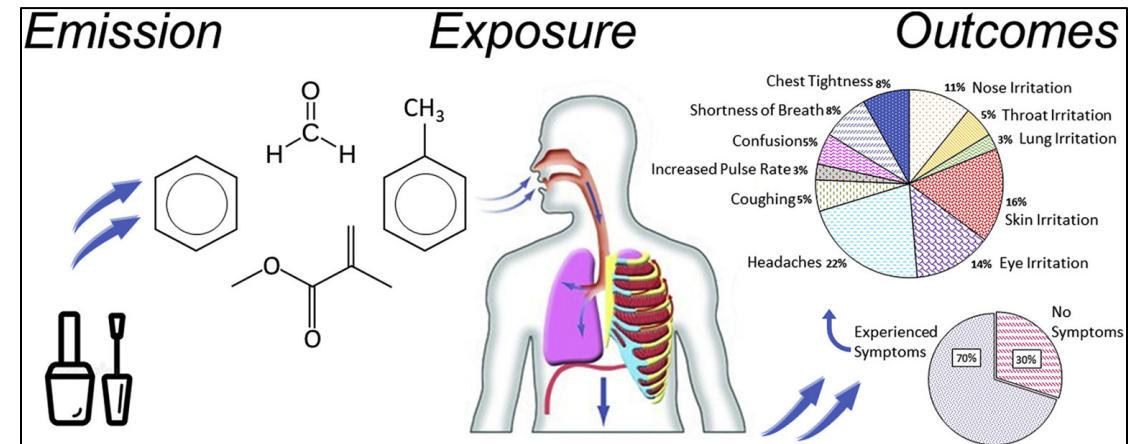
Why do VOC's Matter?

- VOC's can transform into other pollutants (PM and ozone)
- Impacts industry workers that have higher exposure
 - Nail technicians
 - Painters

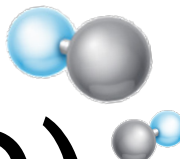


CDPHE hits Western Sugar with \$2 million fine in deal aimed at cleaning up air, land and water at Fort Morgan facility

State officials disclosing settlement say company will overhaul plant



Carbon Monoxide (CO)



Carbon Monoxide (CO)

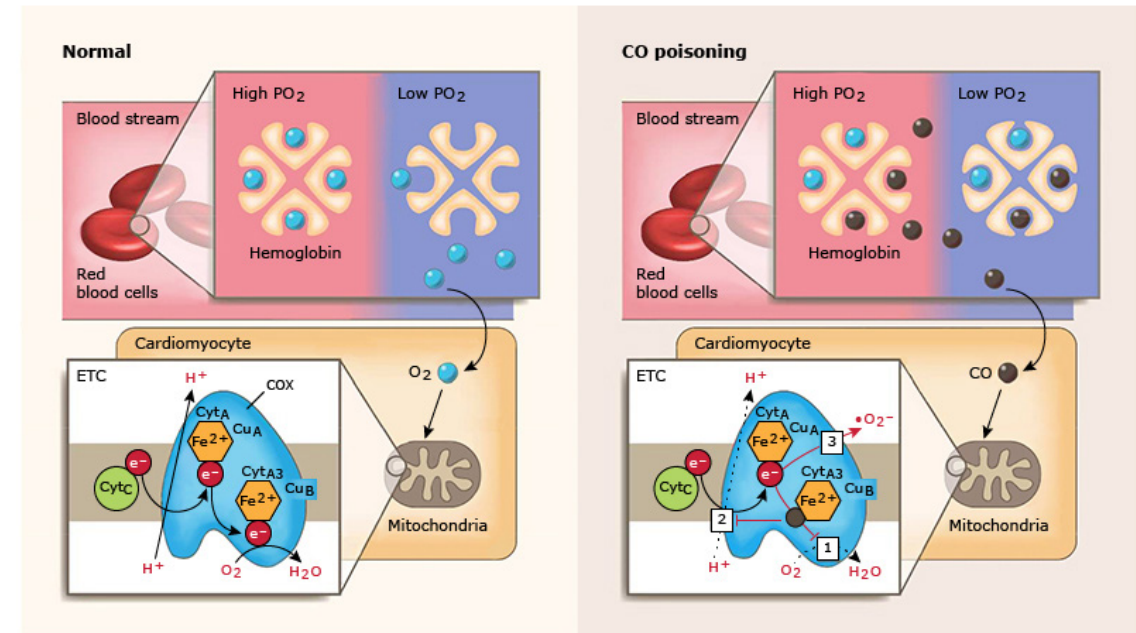
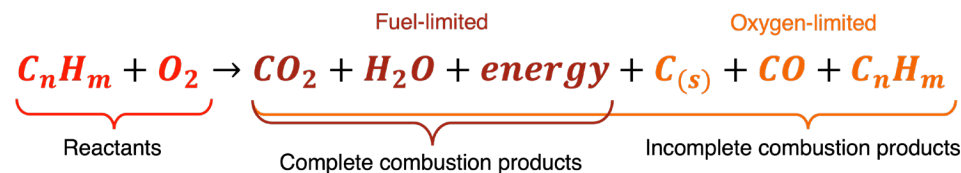


- A colorless, odorless gas formed from **incomplete combustion**

- Motor exhaust
- Industrial manufacturing
- Forest fires

CO is incredibly dangerous!

It can prevent your blood from transporting oxygen



CO & Air Pollution Control!

In 1975, catalytic converters were built to help curb CO emissions from vehicles

Check out this General Motors advertisement from 1975 showing off a new catalytic converter!

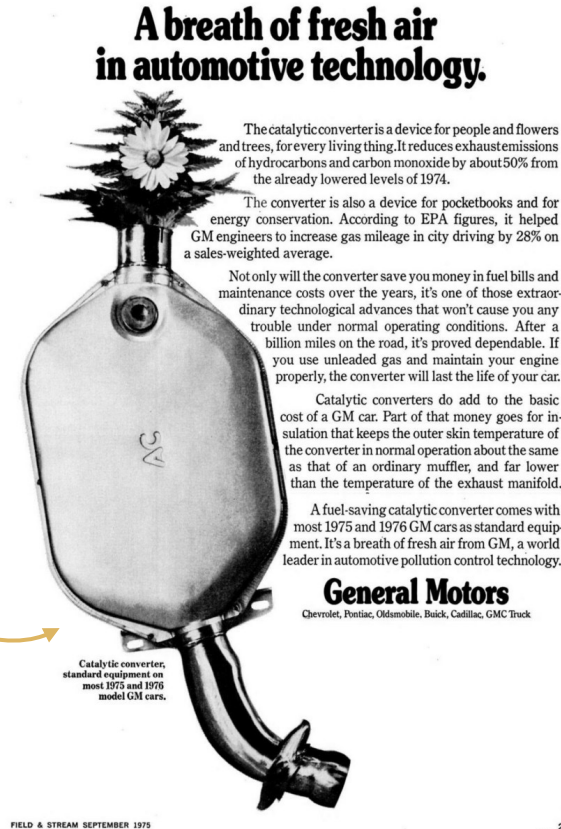
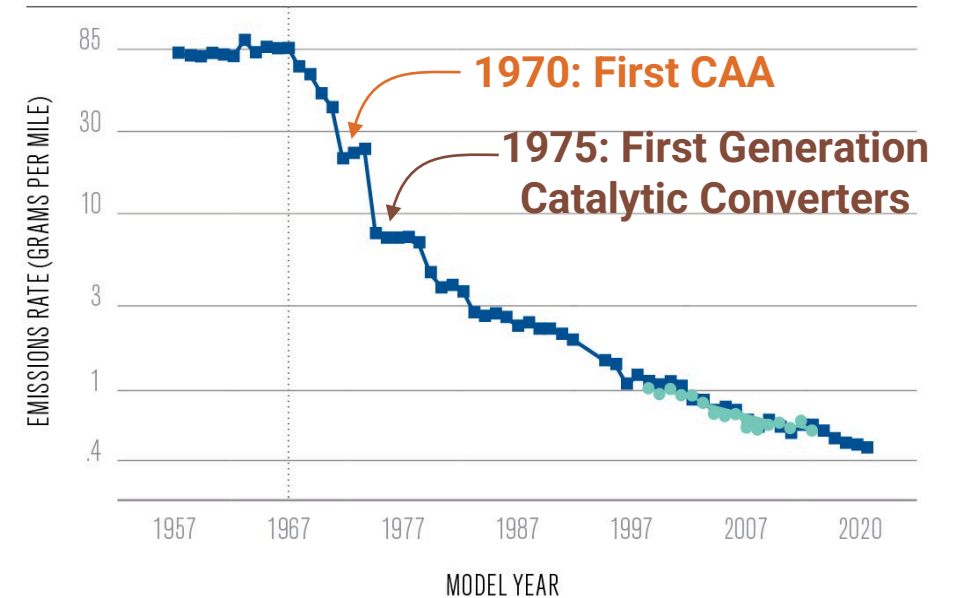


FIGURE 1: MEAN POLLUTION RATES OF NEW U.S. VEHICLES, 1957-2019

(a) Carbon Monoxide (CO)



Next Time...

We'll look a little closer at catalytic converters & CO emissions using our YPOD!



Carbon Monoxide (CO)





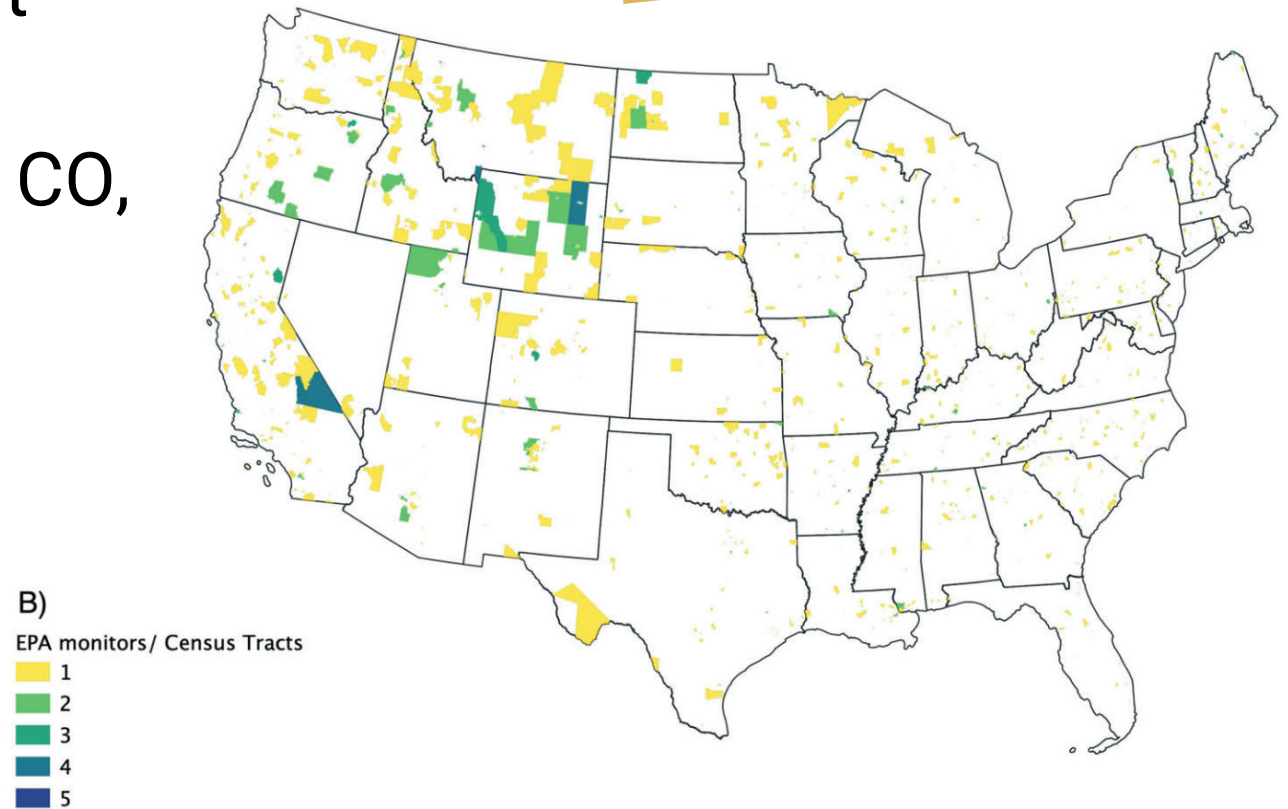
Air Quality Sensing Impacts



Government monitoring

- Highly accurate equipment
 - Expensive!
- Criteria pollutants: PM_{2.5}, CO, NO₂, lead, SO₂, O₃

What do you notice about this map?

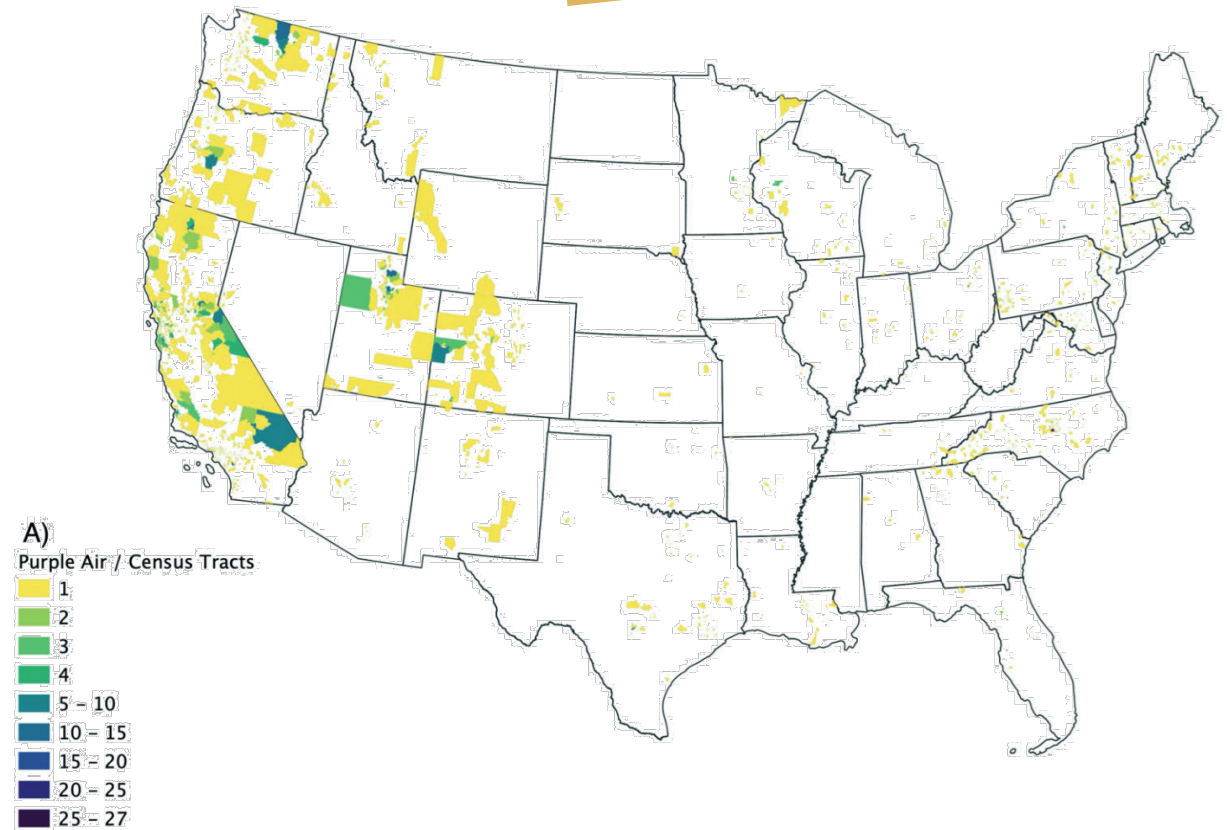


Low-Cost Sensing

- Lower cost in comparison
 - Less accurate
- PurpleAir measures PM 2.5

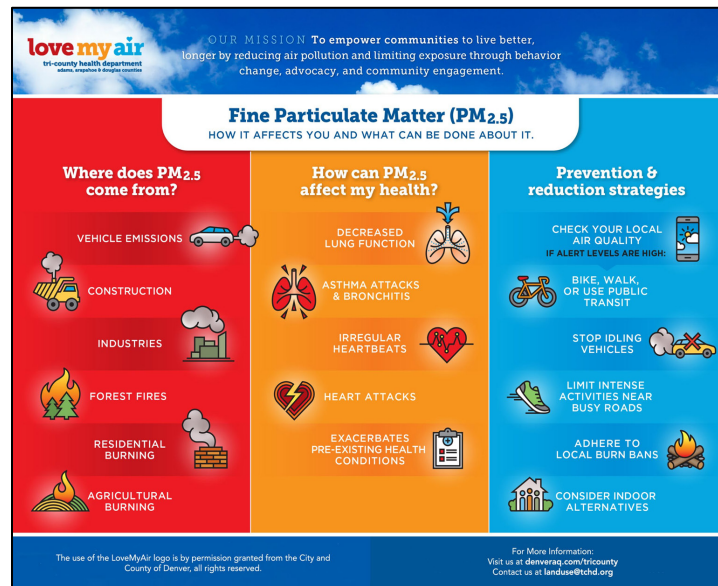


What do you notice about this map?



Low-Cost Sensors and Community Science

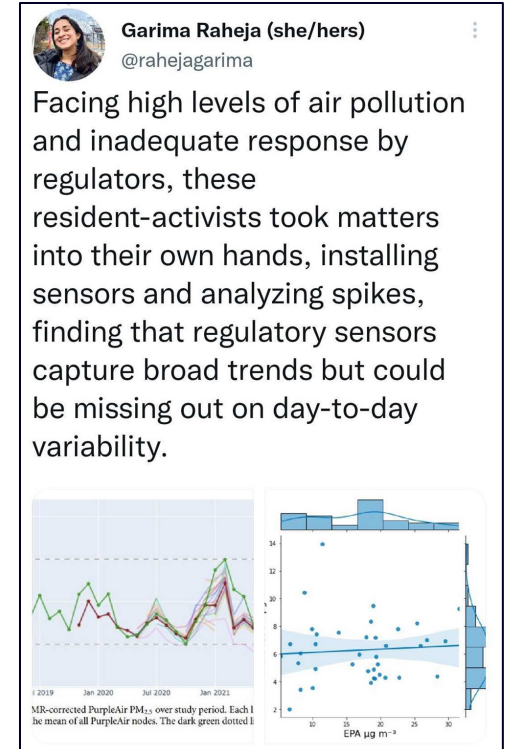
- Support environmental justice initiatives
 - Community members and scientists work together



Love My Air - DDPHE and Denver Public Schools collaboration



Marshall Fire research



Concerned Ohio River Residents grassroots.org



If you want to learn more, we recommend these excellent sources!

Clements, A., Duvall, R., & Greene, D. (2022). *Enhanced Air Sensor Guidebook (2022)*. EPA.
https://cfpub.epa.gov/si/si_public_record_report.cfm?Lab=CEMM&dirEntryId=356426

de Nevers, N. (2017). *Air Pollution Control Engineering* (Third Edition). Waveland Press, Inc.

Polidori, A., Papapostolou, V., Collier-Oxandale, A., Hafner, H., & Blakey, T. (2021). *Community in Action: A Comprehensive Guidebook on Air Quality Sensors*. South Coast Air Quality Management District. South Coast AQMD's AQ-SPEC. <https://www.aqmd.gov/docs/default-source/aq-spec/star-grant/community-in-action-a-comprehensive-guidebook-on-air-quality-sensors.pdf?sfvrsn=10>

